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Master's thesis colloquium
23 October 2026

AI-Based Malaria Diagnosis from Microscopic Thick Blood Smear Images

Resolution robustness of lightweight CNNs for phone-based images

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In cooperation with CarByte Engineering GmbH and EAII

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Presentation Overview

01

Motivation & Project Overview

Why the task matters and how the project is structured

02

Dataset & Methodology

Dataset, preprocessing and CNN architectures

03

Experiments & Results

Architecture and resolution experiments and main results

04

Model Analysis & External Testing

Grad-CAM analysis and external testing

05

Conclusion & Outlook

Main findings, limitations and future work

Motivation

282 million

Estimated malaria cases worldwide in 2024

610,000

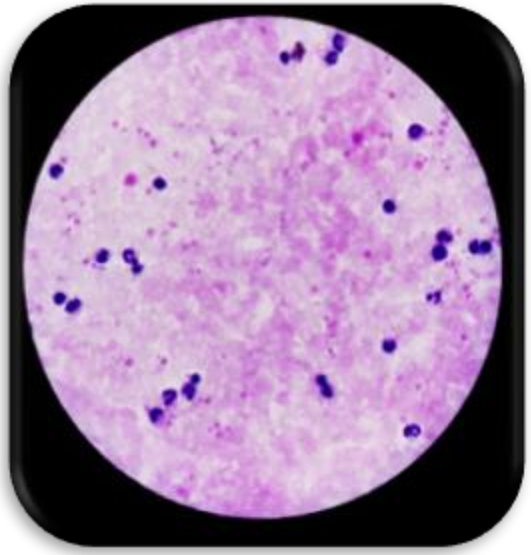
Estimated malaria deaths worldwide in 2024



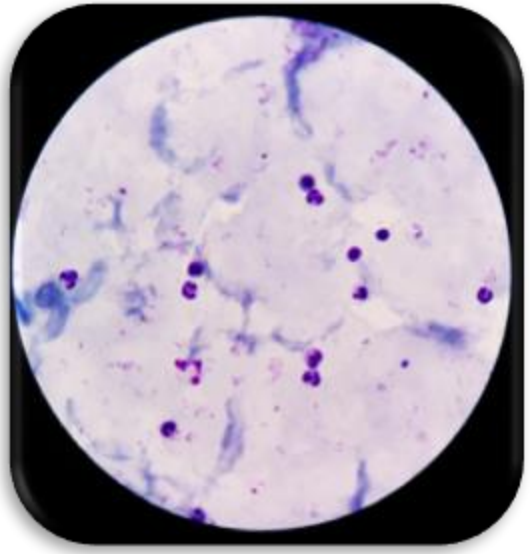
Of malaria deaths occured in the WHO African Region



Plasmodium falciparum



Plasmodium Vivax



Why AI support ?

- Microscopy : Key method for malaria diagnosis
- Manual examination: Requires trained personnel
- Time-consuming and subject to variability

- **AI-assisted prescreening**

Project Overview

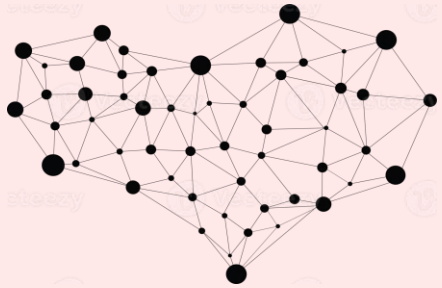
- **Part of an overall project with 3 Master's Projects.**
- Ocular Adapter | Simulation Database | **AI Diagnosis : this thesis**

Input



Thick blood smear images

AI Model



Convolutional Neural Network (e.g. ResNet50)

Output

Uninfected

Infected

Goal
AI-assisted prescreening of malaria microscopy images

Dataset

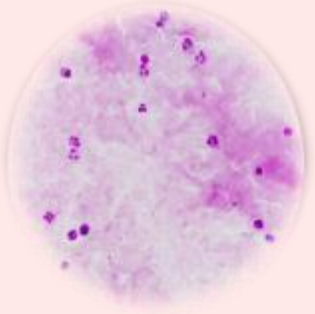
Sources

- NLM (train, val, test)
- AIR Uganda (external test)
- Drassanes–Vall d'Hebron

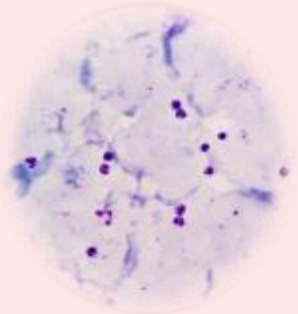
Images

- Giemsa-stained thick blood smears
- Microscopy images

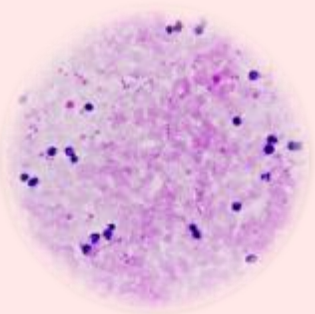
Example Images



Uninfected



Vivax



Falciparum

Classes

- Uninfected
- P. vivax
- P. falciparum

Binary classification

Uninfected: 50 patients

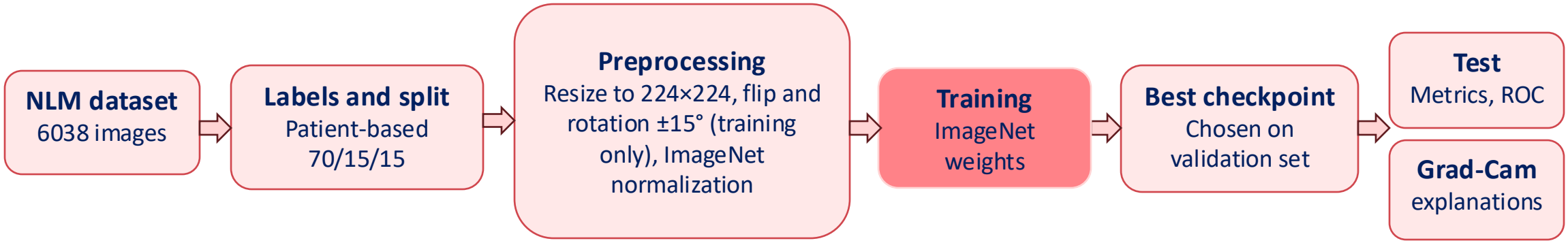
Infected: P. vivax (150 Patients) + P. falciparum (150 patients)

Dataset split

- Patients-based split
- 70% train
- 15% validation
- 15% test

Training and Evaluation Pipeline

One reproducible pipeline, reused for every model and resolution



Training setup

ImageNet pretrained, fine-tuned **15 epochs** **Adam, LR 1e-4(0.0001)** **batch size 8**

Model Architectures

Three models, from accuracy ceiling to mobile deployment

ResNet50

Strong baseline,50-layer residual architecture

Parameters

~ 25.6 M

ResNet18

Smaller ResNet, same design family

Parameters

~ 11.7 M

MobileNetV3-Small

Lightweight architecture, built for mobile devices

Parameters

~ 2.5 M

All three: ImageNet-pretrained, fine-tuned for binary classification. MobileNet-Small received an additional tuning stage.

Resolution Experiments

Original dataset images : 3024 × 4032 px / JPEG and Tiff

HR — Baseline

3024 × 4032

↓

224 × 224

LR1

3024 × 4032

↓

160 × 160

↓

224 × 224

LR2

3024 × 4032

↓

112 × 112

↓

224 × 224

LR3

3024 × 4032

↓

56 × 56

↓

224 × 224

- **Final CNN input:** 224 × 224 for all experiments
- **Only the available image detail is reduced**

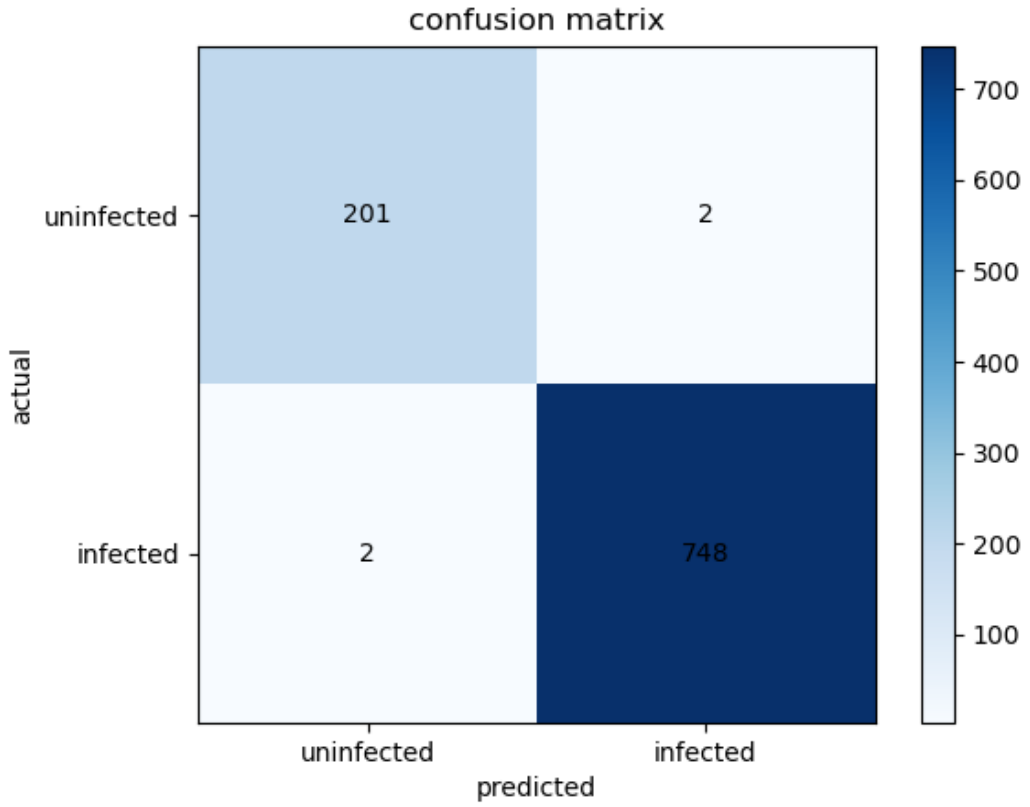
Baseline Results — ResNet50 (224 × 224)

Main Metrics

Accuracy: 99.58%

Sensitivity: 99.73%

Specificity: 99.01%



Test set: 953 images

Only 4 of 953 test images were misclassified

Baseline Results (HR vs LR1 vs LR2 vs LR3)

Resolution Comparison — ResNet50

Resolution	Accuracy	Sensitivity	Specificity
HR 224×224	99.58%	99.73%	99.01%
LR1 160×160	99.58%	99.87%	98.52%
LR2 112×112	98.43%	99.60%	94.09%
LR3 56×56	92.86%	98.00%	73.89%

Key finding:

- Performance stays similarly high at LR1 and LR2, but drops clearly at LR3.
- **Decision:** ResNet18 and MobileNet are trained and tested only at HR and LR3 .

Architecture Comparison

Architecture Comparison at High Resolution (HR)

Model	Accuracy	Sensitivity	Specificity
ResNet50	99.58%	99.73%	99.01%
ResNet18	99.69%	100.00%	98.52%
MobileNetV3-Small	99.69%	100.00%	98.52%

Key finding:

- **All three architectures achieved almost identical performance.**
- **ResNet18 and MobileNetV3-S reached the highest accuracy with no false negatives on the internal test set.**

Architecture Comparison: HR vs LR3

Model	Resolution	Accuracy	Sensitivity	Specificity	ROC-AUC
ResNet50	HR (224px)	99.58%	99.73%	99.01%	0.9997
ResNet50	LR3(56px)	92.86%	98.00%	**73.89%**	0.9698
ResNet18	HR (224px)	99.69%	100.00%	**98.52%**	1.0000
ResNet18	LR3(56px)	91.08%	94.53%	**78.33%**	0.9666
MobilNetV3-S	HR (224px)	99.69%	100.00%	**98.52%**	0.9999
MobilNetV3-S	LR3(56px)	**89.30%**	94.80%	**68.97%**	0.9378

Key finding:

- **All three tied at HR.**
- **At LR3, MobileNet is worst on specificity. ResNet50 keeps the best accuracy and sensitivity, but ResNet18 actually edges it out on specificity**
- **Deployement Model needs optimisation**

Deployment: further model optimisation of MobileNetV3-S

Baseline

Learning rate: 0.0001
Batch size : 8
Epochs: 15
No class weights

First Optimisation

Learning rate: 0.00003
Batch size : 16
Epochs: 30
No class weights

Second Optimisation

Learning rate: 0.00003
Batch size : 16
Epochs: 30
class weights: [2.0, 1.0]

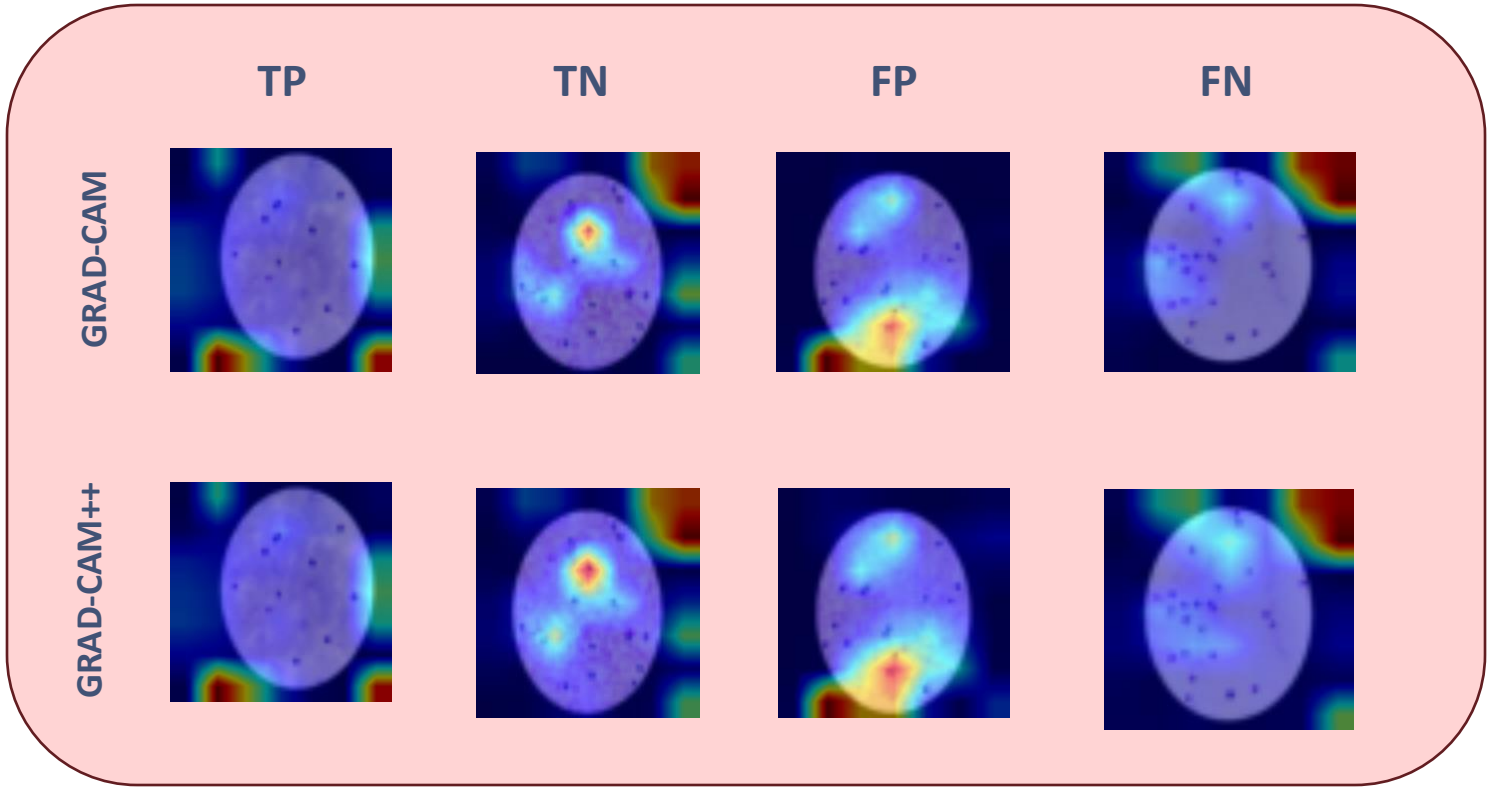
Model	Resolution	Accuracy	Sensitivity	Specificity	ROC-AUC
MobilNetV3-S	LR3(56px)	**89.30%**	94.80%	**68.97%**	0.9378
MobilNetV3-S (opt.1)	LR3(56px)	**92.34%**	99.20%	**67.00%**	0.9711
MobilNetV3-S (opt.2)	LR3(56px)	**92.76%**	96.40%	**79.31%**	0.9506

Key finding:

- **Best trade-off achieved after class weighting:** higher specificity with still high sensitivity.

Grad-CAM and Grad-CAM++ Analysis

ResNet50 at LR3: Strong metrics are not enough, some models also look at the image edges



What it shows:
The image regions that drive each prediction, for TP, TN, FP and FN

What we found
In low-resolution models, hotspots appear on borders and black corners

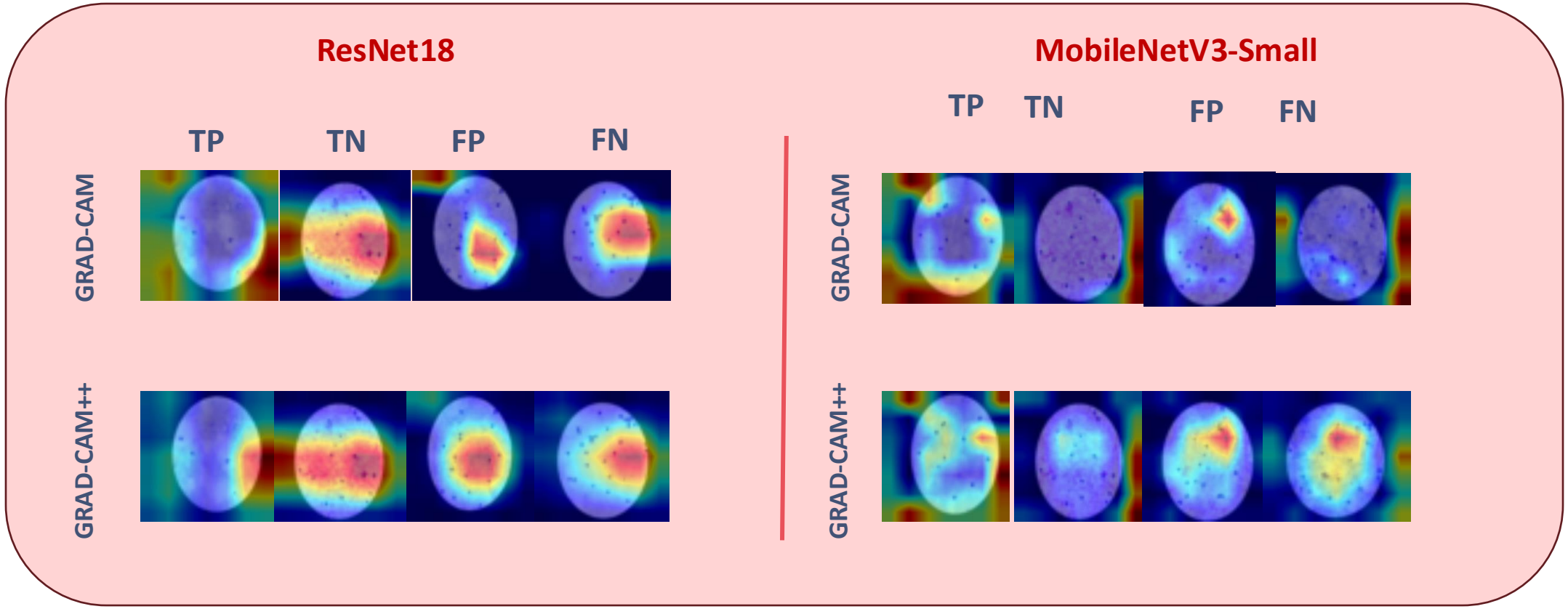
Why it matters
Possible shortcut: the circular field of view instead of the parasites.

Observation:

- Similar activation regions with both methods
- Grad-CAM++ shows stronger / more localized activation

Grad-CAM and Grad-CAM++ Analysis

Explainability Analysis of Smaller Architectures at LR3

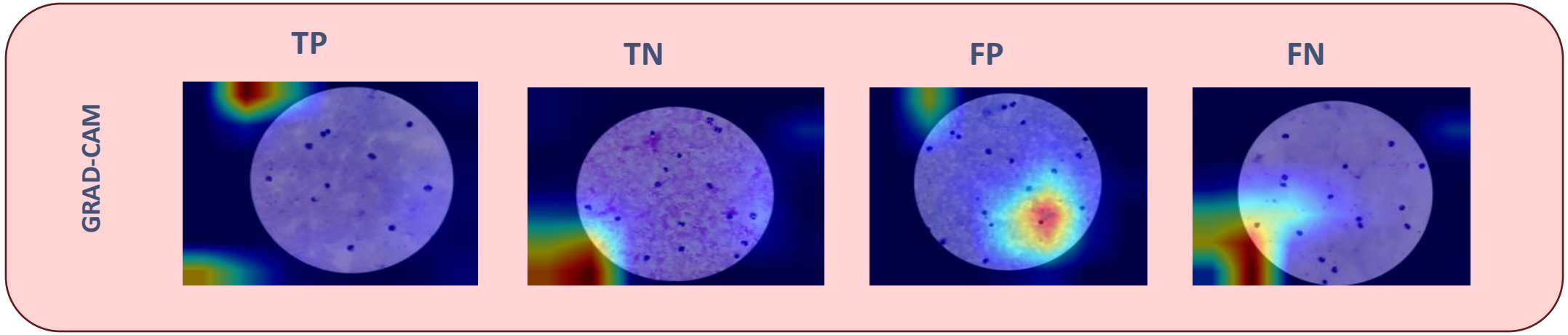


Observation:

- Same behaviour as ResNet50 at HR
- Grad-CAM++ again shows stronger activation

Grad-CAM and Grad-CAM++ Analysis

ResNet50 at HR: Same activation pattern



Observation:

- Similar border / black-corner activation also appears at HR
- Pattern is therefore not limited to LR3
- Suggests possible dataset-specific visual cues

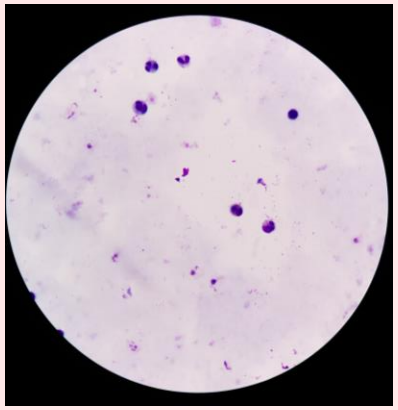
Next step: Test whether the models generalize to an independent dataset

External Validation

Internal test performance is very high, but Grad-CAM suggests possible dataset-specific cues

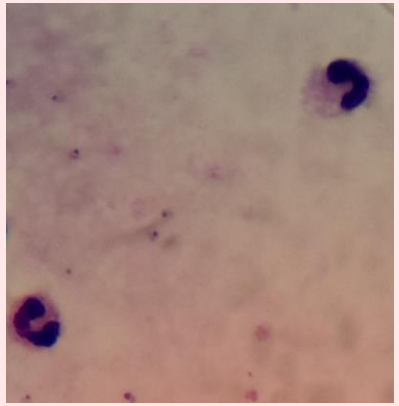
Development Dataset

- NLM/NIH Dataset
- Used for train, val, test



External Dataset

- Independent dataset (AIR Uganda)
- Completely unseen during training



Do the trained models generalize to images from a different dataset?

External Validation : Results

- Uganda phone-camera thick smears (Infected images only).
- 1,182 images: 7,245 annotated parasites.
- No retraining, no tuning: Same preprocessing as each model's own test



- Models still detect parasites on images from a different source.
- Limitation: no uninfected images, so specificity is unknown

Benchmarking

Study	Dataset	Accuracy	Sensitivity	Specificity
This work: MobileNetV3-Small LR3	~6,000 images, 350 patients	92.76%	96.40%	79.31%
Kassim et al. 2021	~6,000 images, 350 patients	83.9% image-level	—	—
Yang et al.	1,819 images, 150 patients	93.46%	92.59%	94.33%
Vásquez Ascate et al.	8,625 images, 400 slides	90.92%	—	—

- **Results are competitive with published work on the same dataset.**
- **Best HR accuracy in this thesis: 99.69%.**

Splits and methods differ between studies: a comparison of performance level, not a strict ranking.

Limitations

Same acquisition : Internal test set is acquired like the training data

Infected-only external set : Sensitivity measured, specificity unknown

File-format confound : Uninfected images are TIFF, infected are JPEG

Binary task only : No species classification, no parasite localization

Not a clinical system : Smartphone deployment and clinical use not yet validated

Conclusion

1. **Very high accuracy**
   ResNet50 reached 99.58% on the internal test set

2. **Robust to resolution loss**
   Performance stayed strong under moderate reduction

3. **Mobile potential**
   MobileNetV3-Small is a candidate for lightweight deployment

4. **Accuracy is not the whole story**
   Grad-CAM and the external test reveal limits to address next

Lightweight CNNs perform strongly

Outlook

Next steps toward a field-ready tool

1. More external datasets, including uninfected images
2. Species classification (P. falciparum / P. vivax)
3. Parasite localization
4. Integration with the smartphone adapter and EAII data
5. Clinical validation

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Thank you for your attention

Questions?

GitHub Repository